Matched-Filtering to Determine Flash Onset based on Neural Responses in the Mouse Visual Cortex

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August 3, 2026

1 Introduction

The realization that the brain can be studied quantitatively as a complex input-output system, where neurons exhibit measurable and sometimes reproducible responses to a stimulus, was foundational to modern neuroscience. This discovery has caused a cascade of further discoveries and innovations, including Brain-computer interfaces (BCI) that can reestablish some communication and movement in paralyzed patients.

This project focused on neural decoding: using a recorded neural response to determine the stimulus presented, a method also common for interfaces. For example, auditory stimulus decoding uses recorded neural activity to determine auditory stimuli, with the potential to be used as a speech neuroprosthetic, enabling direct communication with the brain[1]. A simpler version of neural decoding was explored in this project, wherein a matched-filter was built based on the neural response to flashes, and was later tested to determine whether and when a flash occurred based on recorded neural activity.

The Neuropixels Visual Coding Dataset from the Allen Brain Observatory[2] was used for this project. This data set provides information collected from Neuropixels probes in the visual cortex of C57BL/6 mice from the Brain Observatory 1.1 data set in response to various stimuli, such as flashes, drifting gratings, or natural scenes. This project specifically focused on the neural response to full-field flashes in the primary visual cortex (VISp). The session selected for this project was 732592105, which contained the largest population of VISp neurons.

Disclaimer: AI was used to debug issues with the download of the dataset and to assist with Pandas functions to access data from the dataset.

2 Methods

This project was done in Jupyter across 4 different notebooks:

01_choose_files

02_explore_flashes

03_build_and_validate_matched_filter
04_test_detector

2.1 01_choose_files

This notebook was dedicated to determining the session with the highest number of VISp units. A unit is either a single neuron or a small collection of neurons that are recorded together via the Neuropixels probe. Since this project required determining the population response of neurons in order to reduce noise from a single unit[4], having a higher number of VISp units is expected to result in reduced noise and better predictions. Therefore, the selected session was 732592105, which had 110 VISp units, the largest out of all the sessions.

2.2 02_explore_flashes

This notebook was exploratory, to determine how neurons might respond to flashes. Typically, the onset of a spike occurs 30 to 160 ms after a stimulus[5], and this information was used to bound the region observed around the spike from -0.2 s to 0.6 s. These limits allowed understanding of neural responses prior to the stimulus and encompassed all possible strong responses after stimulus onset. Upon first observation of a single neuron, it did not seem to respond to flash onset. However, after plotting a few more neurons, the response was strongly observable in a few. For example, unit 8 had a strong response at around 50 ms and 300 ms **(Figure 1)**.

To get a clearer picture of how neurons in VISp respond to full-field flashes, **Equation 1** was used to determine the population average of response relative to flash-timing, where R_k is the mean population firing rate in time bin k, and c_{ijk} is the number of spikes from unit j during trial i in time bin k.[3]

$$R_k = \frac{1}{N_{\text{trials}} N_{\text{units}} \Delta t} \sum_{i=1}^{N_{\text{trials}}} \sum_{j=1}^{N_{\text{units}}} c_{ijk} \tag{1}$$

In this exploratory notebook, the population response was observed across 75 -1.0 (black) flashes and across all 110 VISp units **(Figure 2)**

2.3 03_build_and_validate_matched_filter

To determine whether the designed matched filter was accurate, the 150 flash presentations were split according to 75 training flashes to construct the filter, 38 validation flashes to adjust thresholding values to ensure accuracy, and 37 testing flashes to determine whether the filter was accurate. This notebook completed the first two steps: creating the matched filter, and then adjusting its thresholds to increase its accuracy.

The population response for all 110 units across the 75 training flashes was determined based on the method in 02_explore_flashes. In order to normalize the post-stimulus response

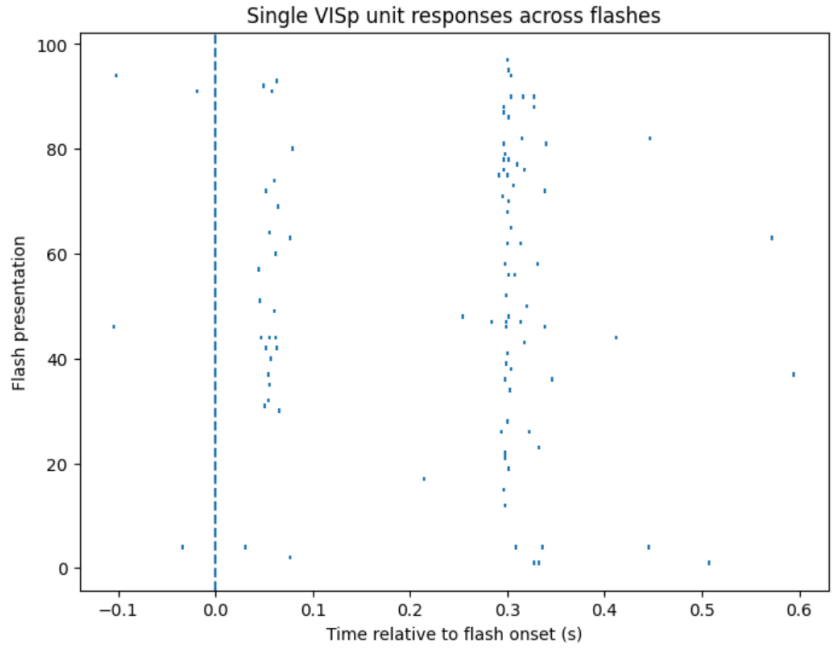


Figure 1: Responses of VISp Unit 8 to flashes. Strong responses observed at around 50 ms and 300 ms.

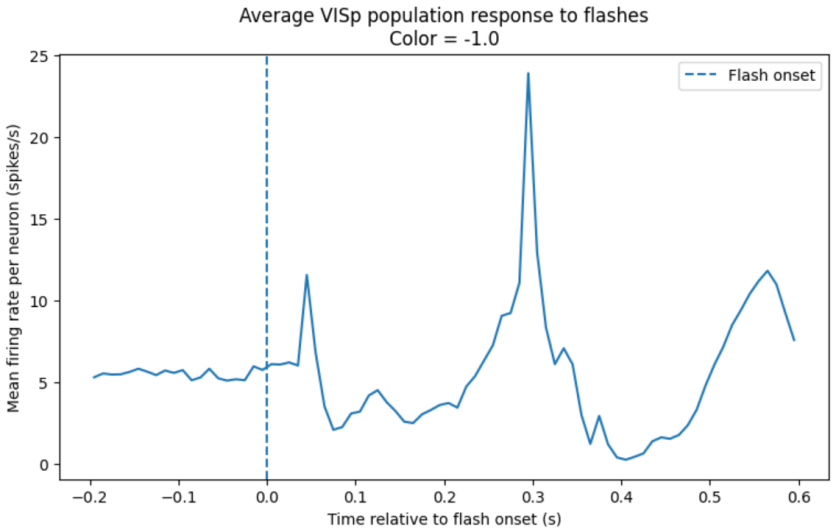


Figure 2: Population Response to Black Flashes

to normal pre-flash activity, the mean firing rate 200 ms before the stimulus was subtracted from the full response. The corrected response from 0 ms to 600 ms after stimulus onset was used as the matched-filter, with the mean subtracted to center the template around 0. These changes are described in **Equation 2**, where R_k is the average population firing rate in time bin k, and $\overline{R}_{\text{post}}$ is the mean firing rate across the selected post-flash interval from 0 to 0.6 s.

$$T_k = R_k - \overline{R}_{\text{post}}, \qquad 0 \leq t_k < 0.6 \text{ s} \tag{2}$$

The population response for all 110 units across the 38 validation flashes was determined based on **Equation 3**. The mean of the data set was subtracted to center the population response of the validation flashes around 0. Importantly, unlike building the template, the response was not averaged across all flash times in order for the cross-correlation to determine when a flash had occurred.

$$R_k^{\text{val}} = \frac{1}{N_{\text{units}}\Delta t} \sum_{j=1}^{N_{\text{units}}} c_{jk}, \qquad x_k = R_k^{\text{val}} - \overline{R^{\text{val}}} \tag{3}$$

Next, the cross-correlation between the template and the population response of the validation flashes, along with their resulting lags was computed using SciPy's signal.correlate function[6]. The 'valid' mode was used so that a correlation score was calculated only when the entire template overlapped with the neural signal to avoid any edge values. The peaks of the cross-correlation were then determined using SciPy's signal.find_peaks function, and plotted **(Figure 3)**. The threshold values of height (required height of peaks) and distance (required distance between neighboring peaks) were adjusted to optimize for the most accurate predictions. The final height was 400 and the final distance was 0.08 ms. Once optimized for the validation set of flashes, the threshold values were fixed, to determine whether this constructed matched filter would be accurate on the testing set of flashes as well.

2.4 04_test_detector

This notebook followed the exact same process as 03_build_and_validate_matched_filter, except the threshold values remained fixed and the template was cross-correlated with the population response of the 110 VISp units to the 37 testing flashes **(Equation 4)**

$$R_k^{\text{test}} = \frac{1}{N_{\text{units}}\Delta t} \sum_{j=1}^{N_{\text{units}}} c_{jk}, \qquad x_k^{\text{test}} = R_k^{\text{test}} - \overline{R^{\text{test}}} \tag{4}$$

3 Results

A prediction was considered correct if it occurred within a certain error tolerance (ms) of an actual flash onset. Of the 37 testing flashes, 33/37 were correctly predicted with a tolerance of 300 ms. The matched-filter had a 51.4% accuracy with a 100 ms error tolerance and

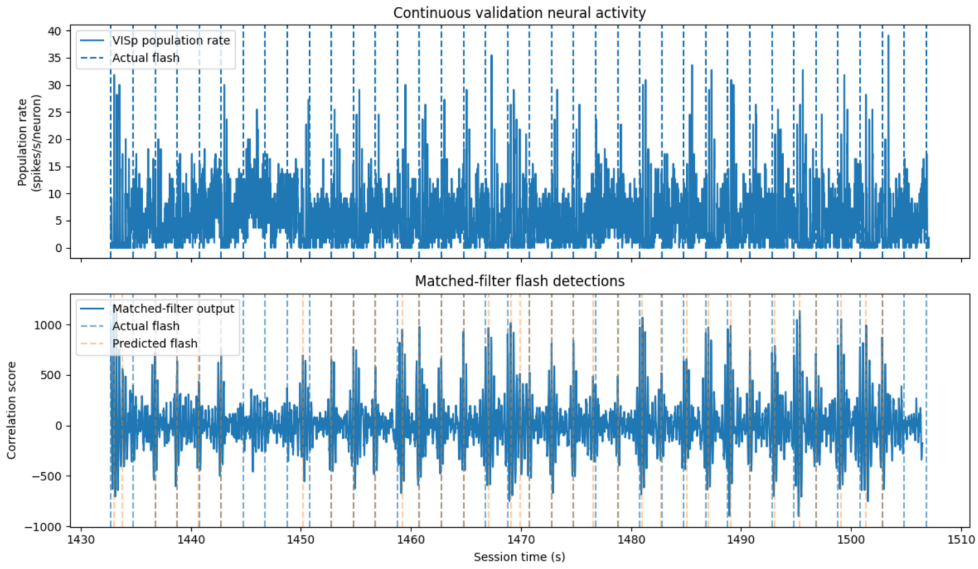


Figure 3: Matched-Filter accuracy for the validation set of flashes. More overlapping orange and blue lines means greater accuracy.

an 89.2% accuracy with a 300 ms error tolerance. Most predicted flashes fell between this 300 ms error range, and increasing the tolerance past this did not result in any significant changes to the accuracy percent. However, a tolerance of 300 ms is relatively broad and the lower performance at 100 ms is indicative of lower temporal precision. The result can be seen in **Figure 4**. Detection errors likely occurred due to a combination of neural variability, spontaneous neural activity that resembled the response to a flash onset, or information loss due to collapsing all individual units into a single population response.

4 Discussion

There were several pivots that had to be made along the way in order to increase the accuracy of the matched-filter. Most notably, this project's initial question was whether the matched-filter could detect flashes of a certain color. This Allen Dataset presented full-field flashes in both -1.0 (black) and 1.0 (white). The project began by detecting only black flashes, which did not work because the population response to both black and white flashes were nearly identical with a few variations **(Figure 5)**. Including all flashes gave both a better template and a more accurate matched-filter. With slightly more advanced techniques, distinguishing between flash colors might be a future step. Other future steps include detecting onset of different stimuli (drifting gratings, natural scenes, etc.) or observing differences in responses to the same stimulus in different brain regions. The Allen dataset has a lot of different variables to be played with, and there are many more interesting projects to be done.

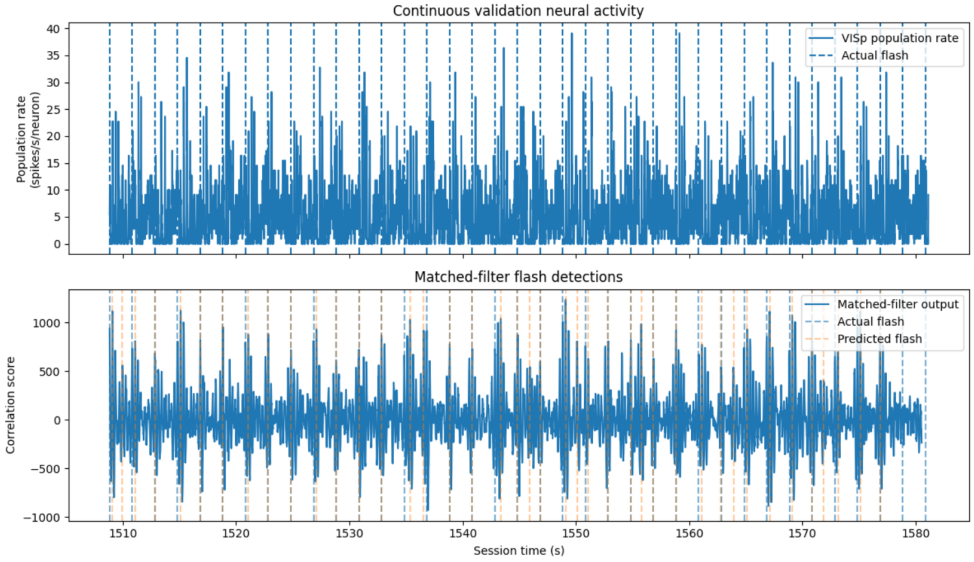


Figure 4: Matched-filter accuracy for the testing set of flashes. An overlap of the orange (predicted flash) and blue (actual flash) means an accurate prediction.

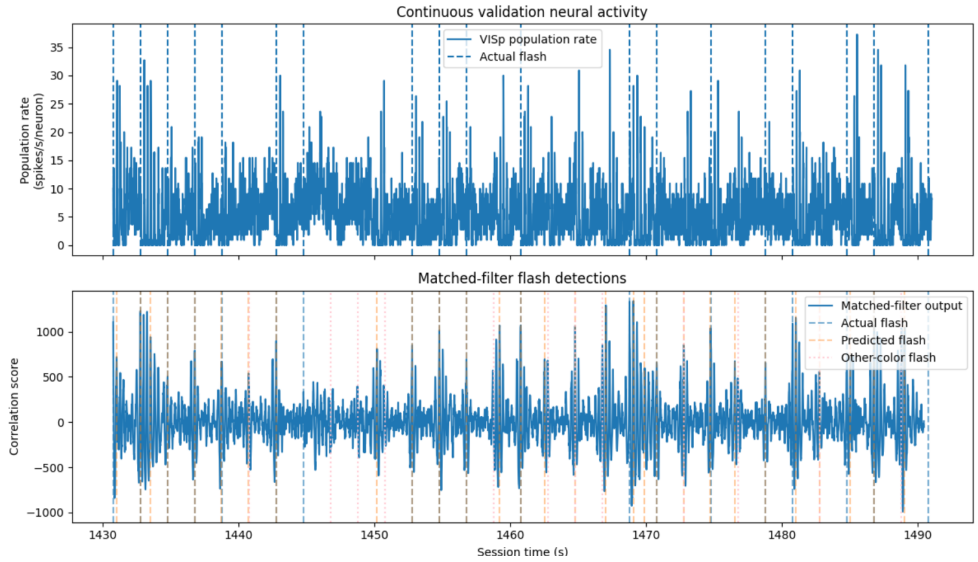


Figure 5: The matched-filter predicted flashes of the other color as well. Overlap of pink (other color flash) and orange (predicted flash) was an indication of this.

References

[1] Hassan Akbari. "Towards reconstructing intelligible speech from the human auditory cortex". In: *Nature Scientific Reports* (2019).

[2] *Accessing Neuropixels Visual Coding Data.*

[3] Gerstner Lab at EPFL. NDC3.1 - Population Coding.

[4] Jorrit S. Montijn. "Population-Level Neural Codes Are Robust to Single-Neuron Variability from a Multidimensional Coding Perspective". In: *Cell Reports* (2016).

[5] Motolani Olarinre. "Relative timing and coupling of neural population bursts in large-scale recordings from multiple neuron populations". In: *bioRxiv Preprint* (2025).

[6] *Signal processing (scipy.signal).*